EXPERIMENT-25 Monkey Banana Problem

Aim

To implement the **Monkey and Banana Problem** using Prolog.

Objectives

- To represent states in a problem-solving environment.
- To use logical rules to determine possible actions.
- To find a sequence of actions for the monkey to obtain the banana.

Algorithm

1. Represent the initial state of the monkey.
2. Define actions such as:
    - Move
    - Push box
    - Climb box
    - Grasp banana
3. Check the preconditions for each action.
4. Generate a sequence that allows the monkey to get the banana.

Flowchart

```
     START
       |
       v
 Monkey at initial position
       |
       v
  Move to box position
       |
```

```
      v
   Push box below
     banana
      |
      v
   Climb onto box
      |
      v
  Reach for banana
      |
      v
 Monkey gets banana
      |
      v
     STOP
```

Program

```
% Monkey Banana Problem

move(monkey, from(door), to(window)).

move(monkey, from(window), to(middle)).

push_box :-
    move(monkey, from(window), to(middle)),
    write('Monkey pushes the box to the middle.'), nl.
```

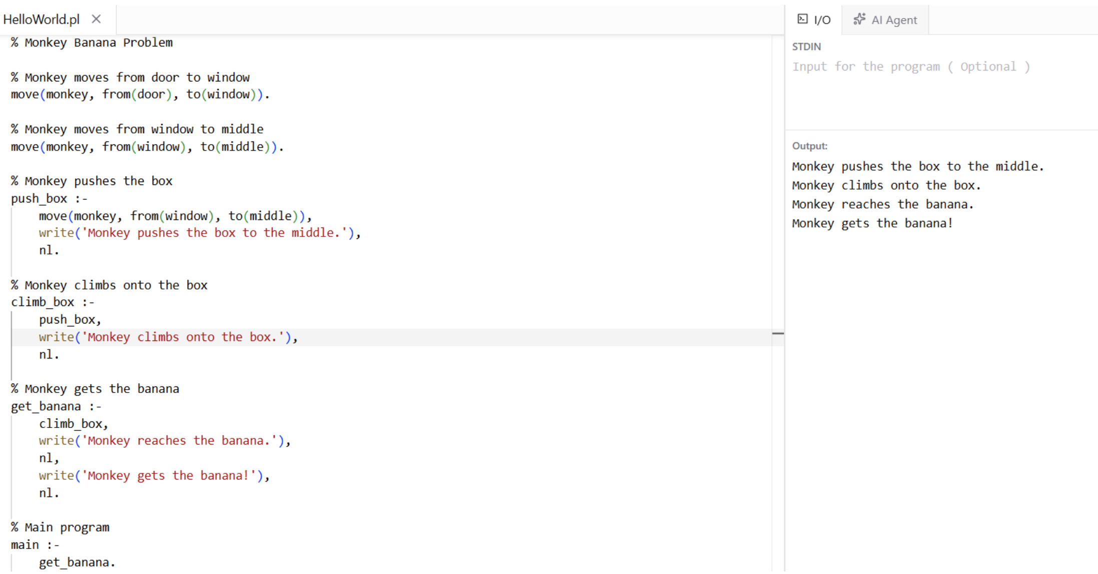
```
climb_box :-
   push_box,
   write('Monkey climbs onto the box.'), nl.


get_banana :-
   climb_box,
   write('Monkey reaches the banana.'), nl,
   write('Monkey gets the banana!'), nl.
```

Query

?- get_banana.

Output



Result

The Monkey and Banana problem was successfully implemented using Prolog rules.

Conclusion

The program demonstrates problem-solving through a sequence of logical actions.